

ST512-SP26-Homework-1

Jason Menjivar-Hernandez

Problem 1

$$n = 36$$

$$\bar{x} = 23.4$$

$$sd(s) = 7.2$$

$$CI: \bar{x} \pm t^* \times SE$$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{7.2}{\sqrt{36}} = 1.2$$

$$t^* = qt(1-(.04/2), 36) = 2.13$$

$$CI: 23.4 \pm 2.13 * 1.2$$

$$96\% CI: (20.84, 25.96)$$

Problem 2

$$H_o : \mu = 3.5$$

$$H_A : \mu < 3.5$$

Will be conducting a left-tailed test because we want to know if the drug delivers relief faster (i.e., less than 3.5 minutes).

$$n = 50$$

$$\bar{x} = 3.1$$

$$s = 1.5$$

$$\alpha = 0.05$$

$$SE = \frac{s}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = 0.212$$

$$t\text{-stat} = \frac{\bar{x} - \mu_o}{SE} = \frac{-0.4}{0.212} = -1.887$$

$$p\text{-value} = \text{pt}(t\text{-value}, df) = 0.033$$

$p\text{-value} < \alpha$, we reject H_o

Therefore, there is sufficient evidence at the 5% confidence interval to conclude that the newly developed pain reliever delivers perceptible pain relief more quickly than the standard pain reliever.

Problem 3

	Design 1	Design 2
n_i	11	6
$\bar{y}_{i\cdot}$	52	46
sd	12	10

a

$$\text{point estimate: } \mu_1 - \mu_2 = \bar{y}_{1\cdot} - \bar{y}_{2\cdot} = 52 - 46 = 6$$

$$95\% \text{ CI: } (\bar{y}_{1\cdot} - \bar{y}_{2\cdot}) \pm t^* \times SE$$

Assuming equal variance:

$$df = n_1 + n_2 - 2 = 15$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{df} = \frac{10(12^2) + 5(10^2)}{15} = 129.33$$

$$s_p = \sqrt{129.33} = 11.37$$

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 11.38 \sqrt{\frac{1}{11} + \frac{1}{6}} = 5.78$$

$$t^* = \text{qt}(0.975, 15) = 2.13$$

$$95\% \text{ CI: } 6 \pm 2.13 * 5.78$$

$$95\% \text{ CI: } (-6.31, 18.31)$$

b

$$H_o : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

$$t\text{-stat} = \frac{\text{pointestimate}-0}{SE} = \frac{6}{5.78} = 1.04$$

$$p\text{-value: } 2 * p(T > |t\text{-value}|) = 0.31$$

$$\alpha = 0.01$$

p-value $> \alpha$; fail to reject H_o

There is not sufficient evidence at the 1% significance level to conclude the mean monthly sale differ between the two designs.

Problem 4

a

This is an experimental design because the treatments are being controlled, the investigator's role is active

b

This is a randomized complete block design.

c

- i. response variable: tomato yield
- ii. factor: fertilizer application rate
- iii. treatments: 3, 5, and 7 units/acre

d

There are 6 replications per treatment.

Problem 5

a

Source	df	Sum of Squares	Mean Squares
Between	2	1533	766.6
Within (Error)	20	6241	312.0
Total	22	7774	

b

Refer to R code at end for these values:

$$F\text{-value} = 2.457$$

$$p\text{-value} = 0.111$$

Our null hypothesis is that the mean phosphorous content does not differ among the three varieties. Using the calculated F-value, 2.457, the p-value was calculated to be 0.111. In comparison to a standard significance level of 5%, our p-value is larger. Therefore, failing to reject the null hypothesis. I think the three varieties have the same mean phosphorous content.

Problem 6

<i>Source of Variation</i>	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F-value</i>	<i>P-value</i>
Between	4	2.281	0.570	4.32	0.0039
With-in (error)	60	7.949	0.132		
Total	64	10.230			

Based on the ANOVA table, the calculated p-value is approximately 0.0039. The p-value is less than the significance level of 0.01, thus we reject the null hypothesis. Therefore, it appears to be a difference in means between the five diets.

Problem 7

a

```
tapply(isoflavones,category,length) #verifies sample sizes
```

```

Cereals      Energy Bars Veggie Burgers
      5              5              5

```

```
tapply(isoflavones,category,mean) # verifies sample means
```

```

Cereals      Energy Bars Veggie Burgers
      9.2             10.0             13.8

```

```
tapply(isoflavones,category,var) # verifies sample variances
```

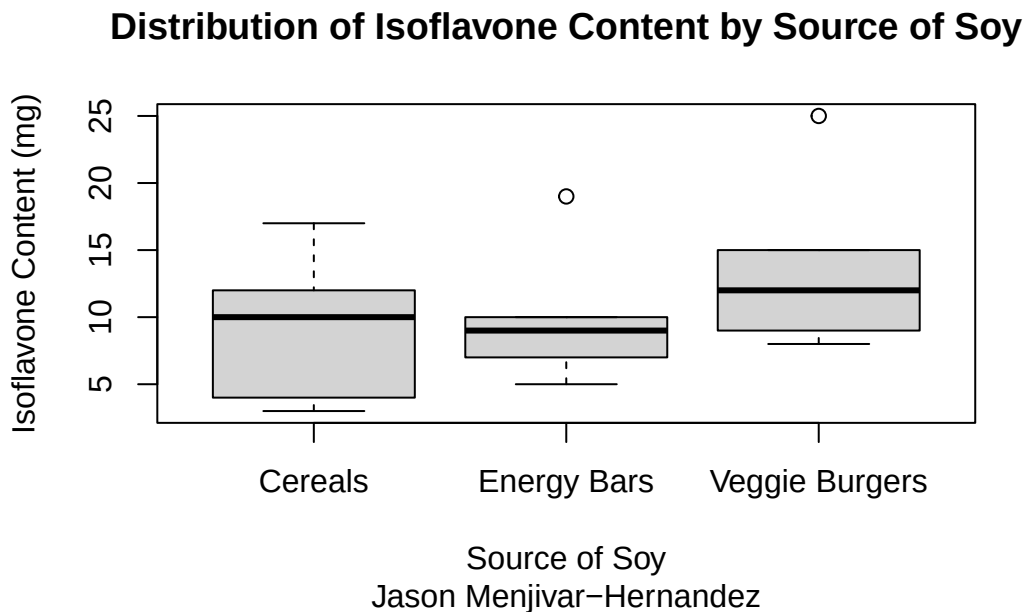
```

Cereals      Energy Bars Veggie Burgers
      33.7             29.0             46.7

```

b

```
boxplot(isoflavones~category, xlab = "Source of Soy", ylab= "Isoflavone Content (mg)", main =
```



Observations:

The cereal data is right-skewed, provided that it had a large upper quartile range. Average isoflavone content due to different source of soy varies (i.e., not equal among the groups). Some of the boxplots appear to be of different sizes, so it is possible that the variances are not the same from group to group, suggesting that the source of soy is associated with isoflavone content.

c

```
model1=aov(isoflavones~category)
summary(model1)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
category	2	60.4	30.20	0.828	0.46
Residuals	12	437.6	36.47		

Based on the above ANOVA table, the p-value is calculated to be 0.46. Using a standard significance level of 0.05, the p-value is larger. Therefore, we fail to reject the null hypothesis. There is no difference in mean isoflavone content between the three sources soy.

R Code

```
qt(1-(.04/2), 36) #problem 1
```

```
[1] 2.130871
```

```
pt(-1.887,49) #problem 2
```

```
[1] 0.03254595
```

```
qt(0.975,15) #problem 3.a
```

```
[1] 2.13145
```

```
2*(1-pt(1.04,15)) #problem 3.b
```

```
[1] 0.314814
```

```
# problem 5 r-code
group <- factor(c(rep("A",8), rep("B",8), rep("C",7)))

value <- c(
  rep(72 + sqrt((8-1)*312/8), 4), rep(72 - sqrt((8-1)*312/8), 4),
  rep(68 + sqrt((8-1)*310/8), 4), rep(68 - sqrt((8-1)*310/8), 4),
  rep(55 + sqrt((7-1)*321/7), 3), rep(55 - sqrt((7-1)*321/7), 4)
)

data <- data.frame(group, value)
anova_model <- aov(value ~ group, data = data)
summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	1533	766.6	2.457	0.111
Residuals	20	6241	312.0		

```
#problem 6 p-value  
1-pf(4.32,4,60)
```

```
[1] 0.003885707
```

```
#problem 7  
isocontent = c(3, 17, 12, 10, 4, 19, 10, 9, 7, 5, 25, 15, 12, 9, 8)  
source = c(rep("1",5),rep("2",5), rep("3",5))  
isoflavone = data.frame(isocontent,source)  
  
group1 <- c(3, 17, 12, 10, 4) # Cereals and Snacks  
group2 <- c(19, 10, 9, 7, 5) # Energy Bars  
group3 <- c(25, 15, 12, 9, 8) # Veggie Burgers  
category <- factor(rep(c("Cereals", "Energy Bars", "Veggie Burgers"), each = 5))  
isoflavones <- c(group1, group2, group3)
```